Insert Interval (medium)

We'll cover the following

Problem Statement

Given a list of non-overlapping intervals sorted by their start time, **insert a given interval at the correct position** and merge all necessary intervals to produce a list that has only mutually exclusive intervals.

Example 1:

```
Input: Intervals=[[1,3], [5,7], [8,12]], New Interval=[4,6]
Output: [[1,3], [4,7], [8,12]]
Explanation: After insertion, since [4,6] overlaps with [5,7], we merged them into one [4,7].
```

Example 2:

```
Input: Intervals=[[1,3], [5,7], [8,12]], New Interval=[4,10]
Output: [[1,3], [4,12]]
Explanation: After insertion, since [4,10] overlaps with [5,7] & [8,12], we merged them into [4,12].
```

Example 3:

```
Input: Intervals=[[2,3],[5,7]], New Interval=[1,4]
Output: [[1,4], [5,7]]
Explanation: After insertion, since [1,4] overlaps with [2,3], we merged them into one [1,4].
```

Try it yourself

Try solving this question here:

Java | Python3 | JS | C++

```
1  import java.util.*;
2
3  class Interval {
```

```
  int start;
  int end;

  public Interval(int start, int end) {
    this.start = start;
    this.end = end;
  }
};

class InsertInterval {

  public static List<Interval> insert(List<Interval> intervals, Interval newInterval) {
    List<Interval> mergedIntervals = new ArrayList<>();
    //TODO: Write your code here
    return mergedIntervals;
  }

  public static void main(String[] args) {
      List<Interval> input = new ArrayList<Interval>();
    input.add(new Interval(1, 3));
    input.add(new Interval(5, 7));
    input.add(new Interval(8, 12));
    System.out.print("Intervals after inserting the new interval: ");
    for (Interval interval : InsertInterval.insert(input, new Interval(4, 6)))
      System.out.print("[" + interval.start + "," + interval.end + "] ");
    System.out.println();

    input = new ArrayList<Interval>();
    input.add(new Interval(1, 3));
    input.add(new Interval(5, 7));
    input.add(new Interval(8, 12));
    System.out.print("Intervals after inserting the new interval: ");
    for (Interval interval : InsertInterval.insert(input, new Interval(4, 10)))
      System.out.print("[" + interval.start + "," + interval.end + "] ");
    System.out.println();

    input = new ArrayList<Interval>();
    input.add(new Interval(2, 3));
    input.add(new Interval(5, 7));
    System.out.print("Intervals after inserting the new interval: ");
    for (Interval interval : InsertInterval.insert(input, new Interval(1, 4)))
      System.out.print("[" + interval.start + "," + interval.end + "] ");
    System.out.println();
  }
}

```

Run Save Reset

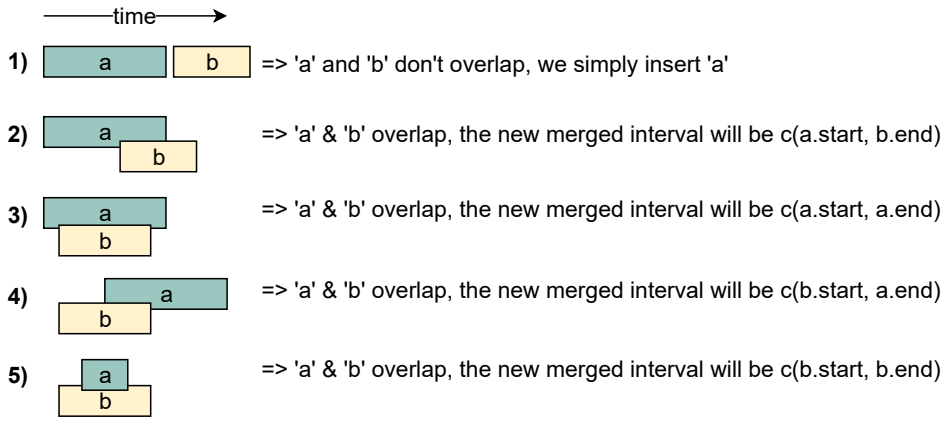
Solution

If the given list was not sorted, we could have simply appended the new interval to it and used the `merge()` function from Merge Intervals. But since the given list is sorted, we should try to come up with a solution better than $O(N * logN)$

When inserting a new interval in a sorted list, we need to first find the correct index where the new interval can be placed. In other words, we need to skip all the intervals which end before the start of the new interval. So we can iterate through the given sorted listed of intervals and skip all the intervals with the following condition:

```
intervals[i].end < newInterval.start
```

Once we have found the correct place, we can follow an approach similar to Merge Intervals to insert and/or merge the new interval. Let's call the new interval 'a' and the first interval with the above condition 'b'. There are five possibilities:



The diagram above clearly shows the merging approach. To handle all four merging scenarios, we need to do something like this:

```
c.start = min(a.start, b.start)
c.end = max(a.end, b.end)
```

Our overall algorithm will look like this:

1. Skip all intervals which end before the start of the new interval, i.e., skip all `intervals` with the following condition:

```
intervals[i].end < newInterval.start
```

2. Let's call the last interval 'b' that does not satisfy the above condition. If 'b' overlaps with the new interval (a) (i.e. `b.start <= a.end`), we need to merge them into a new interval 'c':

```
c.start = min(a.start, b.start)
c.end = max(a.end, b.end)
```

3. We will repeat the above two steps to merge 'c' with the next overlapping interval.

Code

Here is what our algorithm will look like:

Java | Python3 | C++ | JS

```
1  import java.util.*;
2
3  class Interval {
4    int start;
5    int end;
6
7    public Interval(int start, int end) {
8      this.start = start;
9      this.end = end;
10   }
11 };
12
```

```java
class InsertInterval {

  public static List<Interval> insert(List<Interval> intervals, Interval newInterval) {
    if (intervals == null || intervals.isEmpty())
      return Arrays.asList(newInterval);

    List<Interval> mergedIntervals = new ArrayList<>();

    int i = 0;
    // skip (and add to output) all intervals that come before the 'newInterval'
    while (i < intervals.size() && intervals.get(i).end < newInterval.start)
      mergedIntervals.add(intervals.get(i++));

    // merge all intervals that overlap with 'newInterval'
    while (i < intervals.size() && intervals.get(i).start <= newInterval.end) {
      newInterval.start = Math.min(intervals.get(i).start, newInterval.start);
      newInterval.end = Math.max(intervals.get(i).end, newInterval.end);
      i++;
    }

    // insert the newInterval
    mergedIntervals.add(newInterval);

    // add all the remaining intervals to the output
    while (i < intervals.size())
      mergedIntervals.add(intervals.get(i++));

    return mergedIntervals;
  }

  public static void main(String[] args) {
    List<Interval> input = new ArrayList<Interval>();
    input.add(new Interval(1, 3));
    input.add(new Interval(5, 7));
    input.add(new Interval(8, 12));
    System.out.print("Intervals after inserting the new interval: ");
    for (Interval interval : InsertInterval.insert(input, new Interval(4, 6)))
      System.out.print("[" + interval.start + "," + interval.end + "] ");
    System.out.println();

    input = new ArrayList<Interval>();
    input.add(new Interval(1, 3));
    input.add(new Interval(5, 7));
    input.add(new Interval(8, 12));
    System.out.print("Intervals after inserting the new interval: ");
    for (Interval interval : InsertInterval.insert(input, new Interval(4, 10)))
      System.out.print("[" + interval.start + "," + interval.end + "] ");
    System.out.println();

    input = new ArrayList<Interval>();
    input.add(new Interval(2, 3));
    input.add(new Interval(5, 7));
    System.out.print("Intervals after inserting the new interval: ");
    for (Interval interval : InsertInterval.insert(input, new Interval(1, 4)))
      System.out.print("[" + interval.start + "," + interval.end + "] ");
    System.out.println();
  }
}
```

Run Save Reset

Time complexity

As we are iterating through all the intervals only once, the time complexity of the above algorithm is $O(N)$, where 'N' is the total number of intervals.

Space complexity

The space complexity of the above algorithm will be $O(N)$ as we need to return a list containing all the merged intervals.

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Completed Next

Merge Intervals (medium)

Intervals Intersection (medium)